

NLP Assignment 2

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Abstract

This project aims to provide a solution to EDOS Task B, which focuses on categorizing sexist posts into one of the four categories: (1) threats, (2) derogation, (3) animosity, and (4) prejudiced discussion (Kirk et al., 2023). We approached the task using two different pre-trained LLMs, Phi3-mini (3.8B parameters) and Mistral-v3 (7B parameters), and evaluated the models ability to solve this classification task using both zero-shot and few-shot prompting strategies. Metrics such as macro F1 score and fail ratio are computed to assess classification performance and the reliability of model outputs. According to our results, few-shot prompting improves performance compared to zero-shot setting, with both models benefiting from the inclusion of a small number of demonstrations. Among the tested models, Mistral-v3 is most effective demonstrating higher classification capability.

1 Introduction

The Explainable Detection of Online Sexism (EDOS) challenge Task B focuses on assigning sexist posts to one of four categories: threats, derogation, animosity, and prejudiced discussion (Kirk et al., 2023).

There exist many solutions to sexism and offensive language detection. Traditional machine learning models, such as SVMs, logistic regression, or random forests, rely on engineered features like bag-of-words or TF-IDF. These models are lightweight and interpretable but they struggle to capture context or irony. Deep learning approaches, such as LSTMs and transformer-based architectures can model complex contextual patterns and generally achieve higher performance. However, they typically require substantial amounts of labeled data for training or fine-tuning. In this project we propose another approach, which consists in using large language models (LLMs) to perform classification via prompting without the need

of task-specific training. The workflow involves preparing the models and tokenizers, formatting input texts into instruction-style prompts, and optionally injecting few-shot demonstrations to guide the model’s predictions. Generated responses are decoded and mapped to numeric labels, allowing for a quantitative evaluation. Our study shows that, even if LLMs have a great potential, sometimes their outputs can be inconsistent with the prompt directives but few-shot prompting can improve the LLM classification abilities.

2 System description

The system implemented for this assignment consists of a classification pipeline based on LLMs prompting. The pipeline can be summarized as follows: model preparation, prompt construction, generation of predictions, metric computation and error analysis.

Model Preparation We load two pretrained LLMs, Phi3-mini and Mistral-v3, together with their corresponding tokenizers using HuggingFace Transformers (Wolf et al., 2020). We apply quantization, configure device mapping, and set the generation parameters required for inference.

Prompt Construction Two prompting strategies are implemented: zero-shot and few-shot. In the zero-shot setting, each input example is formatted into an instruction-style template. In the few-shot setting, we inject into the prompt also some labeled demonstrations. Labeled demonstrations have been selected randomly or based on semantic similarity. Demonstrations are then selected as the most similar examples to the centroid (referred as *only_similar*), or as a combination of the most and least similar ones (referred as *also_similar*).

Inference During the inference loop we collect all the generated textual responses. Responses are decoded using the associated tokenizer, and a custom parsing function extracts the predicted label from the generated text.

Output Processing and Metrics Predicted labels are mapped into numerical classes and compared to the true labels associated to each example. Performances are evaluated using macro F1 score and fail ratio.

Error Analysis We report results obtained across different numbers of examples included in the prompt and present classification reports and confusion matrices. We also analyze common errors between the two model by printing out the misclassified sentences. Additionally, we analyze how individual words influence the LLM’s classification ability.

3 Experimental setup and results

Both models were run with identical generation settings to ensure a fair comparison. In particular, we set: `temperature = None` to enforce deterministic outputs and `max_new_tokens = 30` to avoid long generations. We evaluated each model both in a zero-shot and few-shot prompting configuration. In particular, for the few-shot configuration, we experimented with different numbers of input examples, testing each model with 2, 3, or 4 demonstration samples. We collected all the responses generated by the models and mapped them to their corresponding numeric labels according to the following mapping: `not-sexist` $\rightarrow$ 0, `threats` $\rightarrow$ 1, `derogation` $\rightarrow$ 2, `animosity` $\rightarrow$ 3, `prejudiced` $\rightarrow$ 4, `model_failure` $\rightarrow$ 5. To evaluate the quality of the model predictions, we used macro F1 score and fail ratio. The fail ratio is defined as the number of predictions assigned to class 5 divided by the total number of predictions. For the computation of macro F1, samples mapped to class 5 were treated as if they belonged to class 0. Results obtained are shown in Table 1.

4 Discussion

Mistral-v3 outperforms Phi3-mini across most prompting configurations, likely due to its larger number of parameters. However, despite achieving better results, Mistral-v3 exhibits a higher fail ratio. A qualitative inspection of the outputs reveals that this behavior is mainly caused by Mistral-v3 predicting multiple classes in a single response. Providing in-context examples through few-shot prompting improves performance for both mod-

Table 1: Macro F1 and fail ratio for each model and prompting strategy

Model	Prompting	Macro F1	Fail Ratio
Phi3-mini	Zero-shot	0.3953	0.0000
Phi3-mini	Few-shots 2	0.4170	0.0000
Phi3-mini	Few-shots 3	0.4582	0.0000
Phi3-mini	Few-shots 4	0.4297	0.0000
Mistral-v3	Zero-shot	0.3717	0.0033
Mistral-v3	Few-shots 2	0.4137	0.0467
Mistral-v3	Few-shots 3	0.4701	0.0300
Mistral-v3	Few-shots 4	0.4874	0.0167

els. This improvement is reflected not only in the macro F1 but also in the single classes F1 scores. An exception is observed for the animosity class in the Phi3-mini model, where F1 does not improve from zero-shot to few-shot prompting. We hypothesize that this behavior is influenced by the specific examples included in the prompt. To further investigate this, we repeated the inference by selecting semantically similar examples for the demonstrations. The results show that, while the F1 score for animosity still decreases, the drop is less pronounced compared to randomly selected examples. Moreover, selecting demonstrations based on semantic similarity (*only_similar*), leads to a slight overall performance improvement with respect to the random selection, while the *also_similar* semantic similarity approach not always leads to significant improvements. A qualitative inspection of the misclassified samples reveals that many of the errors correspond to sentences that are ambiguous or difficult to categorize even for humans, often involving sarcasm or implicit language. Finally, analysis at the single-word level did not reveal any correlation between specific words and the errors made by the models.

5 Conclusion

Large Language Models can be effectively employed to address EDOS Task B; however, their performance strongly depends on careful prompt design and the inclusion of guiding examples (especially if demonstrations are selected according to a sound criterion rather than at random). Among the evaluated models, Mistral-v3 consistently outperforms Phi3-mini. Possible future improvements can include a better design of the prompt or a smarter selection of the demonstrations to use in the few-shot setting.

References

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